



NORTH SYDNEY BOYS HIGH SCHOOL

2024 Year 11 Assessment Task 3

Mathematics Advanced

General Instructions

- Reading time - **5 minutes**
- Working time – **90 minutes**
- Write using blue or black pen
- Diagrams may be drawn in pencil
- NSBHS approved calculators may be used
- All necessary working should be shown in every question
- Answer the multiple choice questions **Q1-Q5** on the sheet provided
- Answer **Q6- Q27** in the space provided.
- Marks may not be awarded for carelessly or badly arranged work.

Class Teacher:

- ☐ Ms Lee
- ☐ Mr Berry
- ☐ Mr Umakanthan
- ☐ Dr Vranešević
- ☐ Mrs Moss
- ☐ Mr Ireland
- ☐ Ms Cai/Mr Lott
- ☐ Mrs Sarofim

Student Number: _____

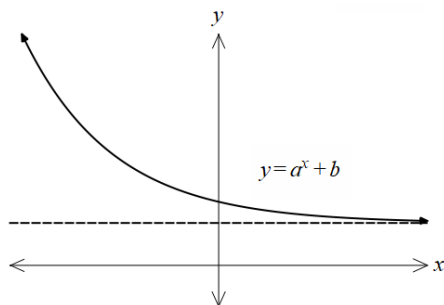
Question No	1-5	6-11	12-15	16-19	20-22	23-25	26-27	Total	Total %
	<u>5</u>	<u>16</u>	<u>14</u>	<u>9</u>	<u>7</u>	<u>7</u>	<u>7</u>	<u>65</u>	<u>100</u>

SECTION I

5 marks Attempt Questions 1–5 Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

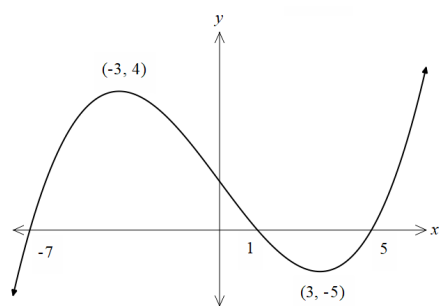
- | Question | Mark | Outcome |
|--|------|---------|
| 1. What is the value of x such that $\log_x 2 = \frac{1}{2}$ | 1 | MA11-6 |
| <p>A. $\sqrt{2}$</p> <p>B. 2</p> <p>C. 4</p> <p>D. $2\sqrt{2}$</p> | | |
| 2. The figure shows the graph of $y = a^x + b$, where a and b are constants. Which of the following must be true? | 1 | MA11-6 |



- A. $0 < a < 1$ and $b > 0$
- B. $0 < a < 1$ and $b < 0$
- C. $a > 1$ and $b > 0$
- D. $a > 1$ and $b < 0$
- | | | |
|--|---|--------|
| 3. What is the derivative with respect to x of $\frac{1}{(3x-4)^5}$ | 1 | MA11-5 |
| <p>A. $\frac{-15x}{(3x-4)^6}$</p> <p>B. $\frac{-15}{(3x-4)^6}$</p> <p>C. $\frac{-5}{(3x-4)^6}$</p> <p>D. $\frac{4}{3(3x-4)^4}$</p> | | |

4. The graph of $y = f(x)$ shown below.

1 MA11-5



For what values of x is $f(x) < 0$ and $f'(x) < 0$?

- A. $(-\infty, -7) \cup (1, 5)$
- B. $(-\infty, -3) \cup (3, \infty)$
- C. $(-3, 3)$
- D. $(1, 3)$

5.

1 MA11-2

The function $g(x)$ and $h(x)$ are defined as follows:

$$g(x) = \frac{x^2}{8}$$

$$h(x) = \sqrt{2x} + 3$$

Which is the correct expression for $h(g(x))$?

- A. $\frac{x}{2} + 3$
- B. $\frac{x}{2} - 3$
- C. $\sqrt{\frac{2x}{3}}$
- D. $\sqrt{\frac{x}{2} + 3}$

END OF SECTION I

Section II**65 marks****Attempt Questions 6-27****Allow about 1 hours and 22 minutes for this section**

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet on pages 21 - 24. If you use this space, clearly indicate which question you are answering.

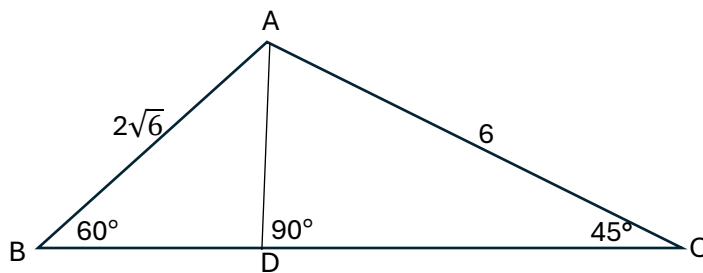
Question	Mark	Outcome
6. Solve the equation $\left \frac{x}{2} - 3\right = 1$	2	MA11-1
7. Simplify $\frac{12^x \times 18^x}{3^x \times 2^x}$	2	MA11-1

8. Find the centre and radius of the circle represented by the
Equation: $x^2 - 6x + y^2 + 10y + 18 = 0$

2 MA11-2

9.

MA11-3



In the diagram, ABC is triangle in which $AB = 2\sqrt{6}$, $AC = 6$, $\angle ABC = 60^\circ$ and $\angle ACB = 45^\circ$. AD is perpendicular from A to BC .

a.

Show that $BC = \sqrt{6} + 3\sqrt{2}$

2

b.

Hence show that $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$

2

10.

- a. In the space below, sketch the graph of the function $y = \sqrt{4 - x}$ showing the intercepts on the axis.

2 MA11-2

- b. State the domain and the range of the function $y = \sqrt{4 - x}$.

2 MA11-2

- 11.** ABC is a triangle in which $\tan B = \frac{1}{3}$. Find the exact value of $\sec B$.

2 MA11-3

12. Differentiate the following expressions with respect to x .

MA11-5

a. $(6x^2 - 5x)^5$

1

b. $7x^2 + 4 - \frac{7}{x}$

1

c. $\frac{3x^3+5}{x-9}$

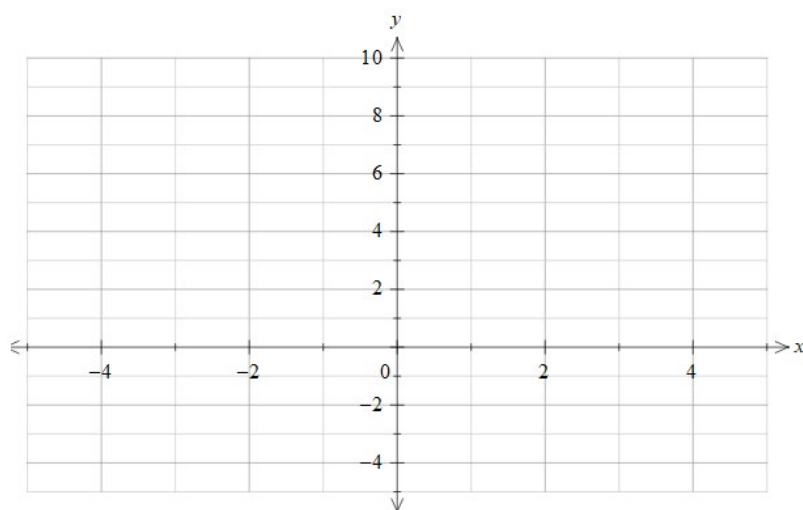
2

- 13.** Find the coordinates of the points on the curve $y = (x^2 - 1)^3$ where the tangent to the curve is parallel to the x - *axis*. **2** **MA11-5**

14. Sketch $y = |2x - 6|$ below.

1 MA11-2

a.



- b. Hence, determine the values of k for which $x + k = |2x - 6|$ has exactly two solutions.

2 MA11-2

15. Given $y = 3x^2 + 6x + 1$. **2** **MA11-5**

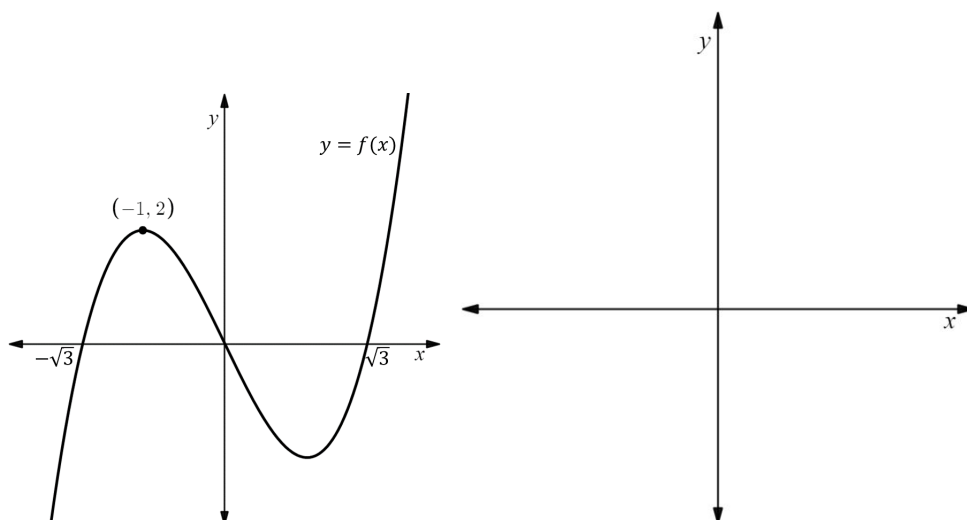
a. Find the equation of the normal at the point (2,25)

b. Find the point A where the normal cuts the x-axis and the point B where the normal cuts y-axis. **2** **MA11-5**

c. If O is the origin find the area of ΔAOB . **1** **MA11-5**

16.

2 MA11-5



The diagram above on the left shows the graph of the curve $y = f(x)$. On the set of axes above on the right, sketch the graph of the curve $y = f'(x)$ showing the shape of the curve and its x intercepts.

17. Prove:

2 MA11-4

$$(1 + \cot^2 \theta) \sqrt{1 - \cos^2 \theta} = \operatorname{cosec} \theta$$

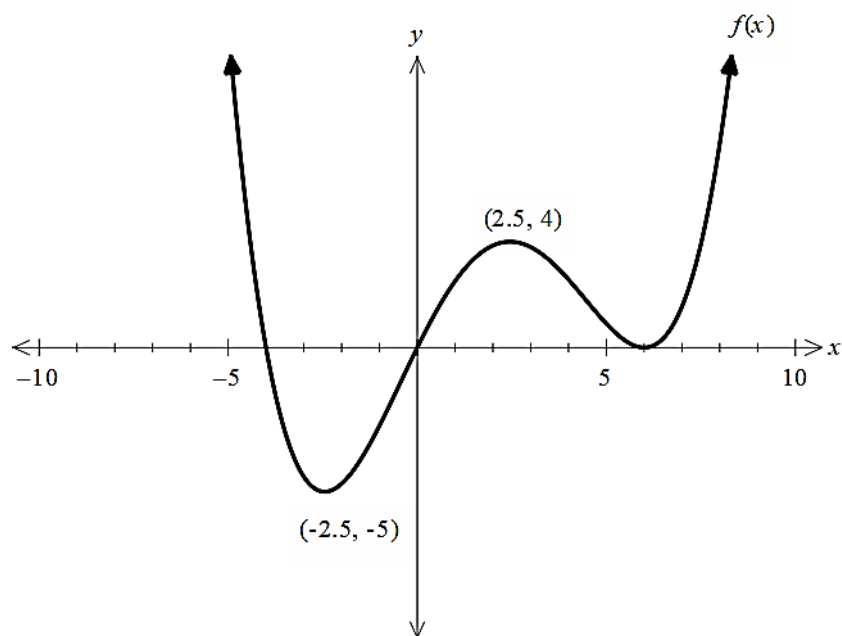
18.

2 MA11-4

$$\text{Solve } (\sqrt{3} \tan \theta - 1)(\sqrt{2} \sin \theta + 1) = 0 \text{ for } -180^\circ \leq \theta \leq 180^\circ.$$

19. The graph of $f(x)$ is shown.

3 MA11-2



Graph $y = 2 - f(x - 1)$ on the same set of axes showing all essential points.

- 20.** The observed brightness of a star can be measured by its energy flux and can be transformed by a logarithmic scale to be more easily measured by an apparent stellar magnitude.

3 MA11-6

It is defined to be $m = -\sqrt[5]{100} \log\left(\frac{f}{f_v}\right)$.

Where m is the magnitude, f is the energy flux of the star and f_v is the reference energy flux.

The star Betelgeuse has a magnitude of 0.45 while the star Deneb has a magnitude of 1.26.

Which star is brighter and by how much brighter is it?

21. Solve the equation $(\log_3 x)^2 = 1$

2 MA11-6

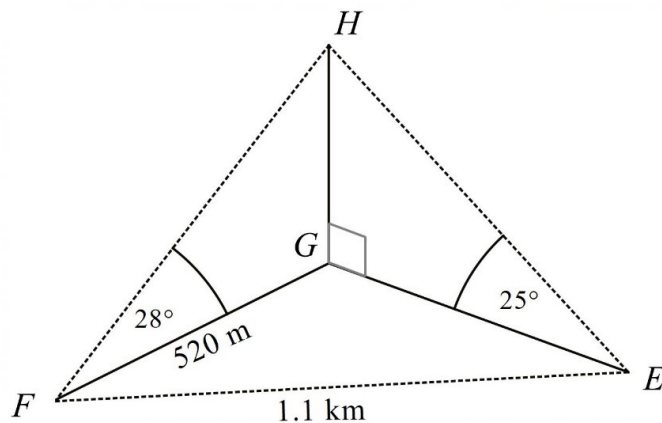
22. Solve the equation $\log_2(x + 4) + \log_2(x - 2) = 4$.

3 MA11-6

- 23.** In the space below, sketch the curve $y = \log_e(x + 1)$ showing any intercepts on the axes and the equation of any asymptote. **2 MA11-6**

- 24.** ABC is a triangle in which $\sin(A + B) = 1$ and $\cos(A - B) = \frac{1}{2}$. Find the sizes of angles A and B in radian given that $A > B$. **2 MA11-4**

25. Three points E , F and G lie on a horizontal plain. The distance $EF = 1.1$ km and $FG = 520$ m. A vertical tower HG is constructed so that $\angle HFG = 28^\circ$ and $\angle HEG = 25^\circ$.



NOT TO
SCALE

Calculate the height of the tower HG and hence find the distance EG and the size of $\angle EGF$, rounding each appropriately.

- 26.** A function $f(x)$ with range $(0, \infty)$ is such that $f(a) \times f(b) = f(a + b)$ for any real numbers a and b .

2 MA11-1

Show that $f\left(\frac{a}{2}\right) = \sqrt{f(a)}$ for any real number a .

27. Nick constructs a swimming pool in the shape below.
 Nick's pool design is made up of a sector and a semi-circle,
 the radius(r) of the sector is equal to the diameter of the semi-circle.

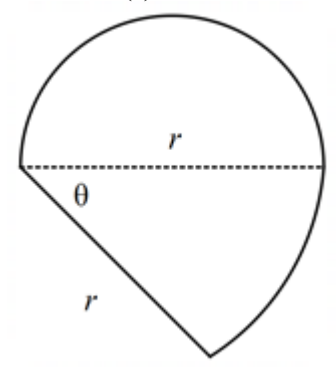


Figure not to scale

- a. Find the perimeter of the pool in terms of r and θ .

1 MA11-3

Nick wants to be able to use this design in several different properties over the next year, so he decides to let the area of the pool equal 1 square unit for future ease in scaling the pool sizes.

- b. Show that, if this is the case, the perimeter can be given by:

$$P = \frac{2}{r} + r \left(1 + \frac{\pi}{4} \right).$$

END OF EXAM



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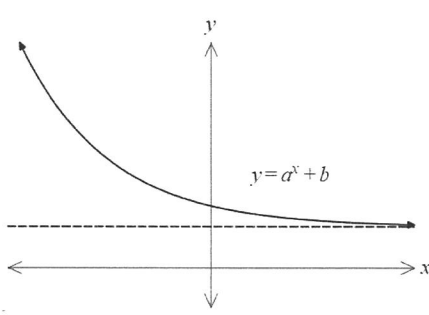
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| <p>2. The figure shows the graph of $y = a^x + b$, where a and b are constants. Which of the following must be true?</p> <div style="text-align: center;">  </div> <p>A. $0 < a < 1$ and $b > 0$</p> <p>B. $0 < a < 1$ and $b < 0$</p> <p>C. $a > 1$ and $b > 0$</p> <p>D. $a > 1$ and $b < 0$</p> | 1 | MA11-6 |
| <p>3. What is the derivative with respect to x of $\frac{1}{(3x-4)^5}$</p> <p>A. $\frac{-15x}{(3x-4)^6}$</p> <p>B. $\frac{-15}{(3x-4)^6}$</p> <p>C. $\frac{-5}{(3x-4)^6}$</p> <p>D. $\frac{4}{3(3x-4)^4}$</p> | 1 | MA11-5 |

Section II

65 marks

Attempt Questions 6-27

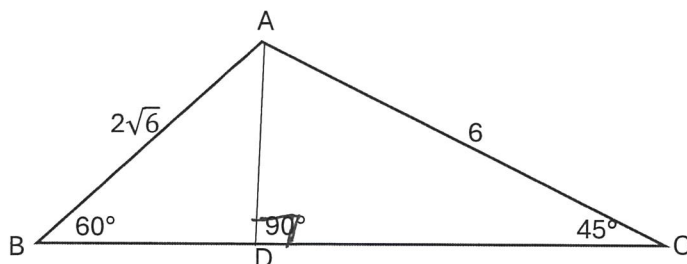
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Question	Mark	Outcome
6. Solve the equation $\left \frac{x}{2} - 3\right = 1$ $\frac{x}{2} - 3 = 1$ $\frac{x}{2} = 4$ $x = 8$ $\frac{x}{2} - 3 = -1$ $\frac{x}{2} = 2$ $x = 4$	2	MA11-1
7. Simplify $\frac{12^x \times 18^x}{3^x \times 2^x}$ $= \frac{(2^2 \times 3)^x \times (3^2 \times 2)^x}{3^x \times 2^x}$ $= \frac{2^{2x} \times 3^{2x} \times 3^{2x} \times 2^{2x}}{3^x \times 2^x}$ $= 2^{2x} \times 3^{2x}$ $= (6)^{2x}$	2	MA11-1

9.

MA11-3



In the diagram, ABC is triangle in which $AB = 2\sqrt{6}$, $AC = 6$, $\angle ABC = 60^\circ$ and $\angle ACB = 45^\circ$. AD is perpendicular from A to BC .

- a. Show that $BC = \sqrt{6} + 3\sqrt{2}$

2

$$\cos 60 = \frac{BD}{2\sqrt{6}} \Rightarrow BD = 2\sqrt{6} \times \frac{1}{2} = \sqrt{6}$$

$$\frac{DC}{6} = \cos 45 \quad DC = 6 \cos 45$$

$$DC = \frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{6\sqrt{2}}{2}$$

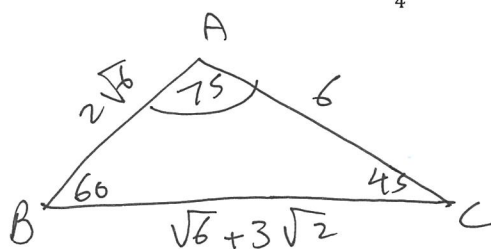
$$DC = 3\sqrt{2}$$

$$\therefore BC = BD + DC$$

$$= \sqrt{6} + 3\sqrt{2}$$

- b. Hence show that $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$

2



$$\frac{\sin 75}{\sqrt{6} + 3\sqrt{2}} = \frac{\sin 60}{6}$$

$$\sin 75 = \frac{\frac{\sqrt{3}}{2}(\sqrt{6} + 3\sqrt{2})}{6}$$

$$\frac{\sin 75}{\sqrt{6} + 3\sqrt{2}} = \frac{\sin 45}{2\sqrt{6}}$$

$$\sin 75 = \frac{\frac{1}{\sqrt{2}}(\sqrt{6} + 3\sqrt{2})}{2\sqrt{6}}$$

$$= \frac{\frac{3\sqrt{2}}{2} + \frac{3\sqrt{6}}{2}}{6}$$

$$= \frac{3\sqrt{2} + 3\sqrt{6}}{12}$$

$$= \frac{3(\sqrt{2} + \sqrt{6})}{12}$$

$$= \frac{\frac{\sqrt{6}}{\sqrt{2}} + 3}{2\sqrt{6}} = \frac{\sqrt{3} + 3}{2\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{3\sqrt{2} + 3\sqrt{6}}{12}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

12. Differentiate the following expressions with respect to x .

MA11-5

a. $(6x^2 - 5x)^5$

1

$$\frac{dy}{dx} = 5(6x^2 - 5x)^4 \cdot x(12x - 5)$$

$$= 5(12x - 5)(6x^2 - 5x)^4$$

b. $7x^2 + 4 - \frac{7}{x}$

1

$$y = 7x^2 + 4 - 7x^{-1}$$

$$\frac{dy}{dx} = 14x + \frac{7}{x^2}$$

c. $\frac{3x^3 + 5}{x - 9}$

2

$$u = 3x^3 + 5 \quad v = x - 9$$

$$u' = 9x^2 \quad v' = 1$$

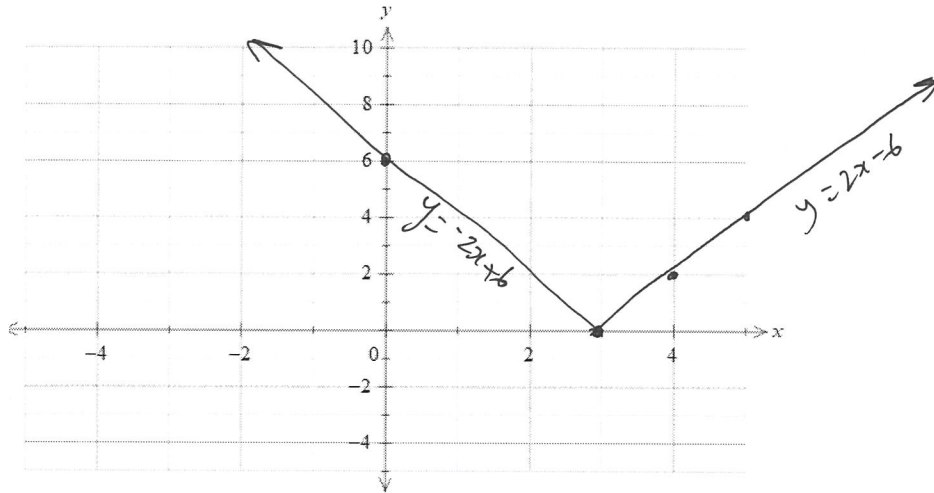
$$\frac{dy}{dx} = \frac{9x^2(x - 9) - (3x^3 + 5)}{(x - 9)^2}$$

$$= \frac{6x^3 - 81x^2 - 5}{(x - 9)^2}$$

14. Sketch $y = |2x - 6|$ below.

1 MA11-2

a.



- b. Hence, determine the values of k for which $x + k = |2x - 6|$ has exactly two solutions.

2 MA11-2

at $(3, 0)$ where $y = |2x - 6|$

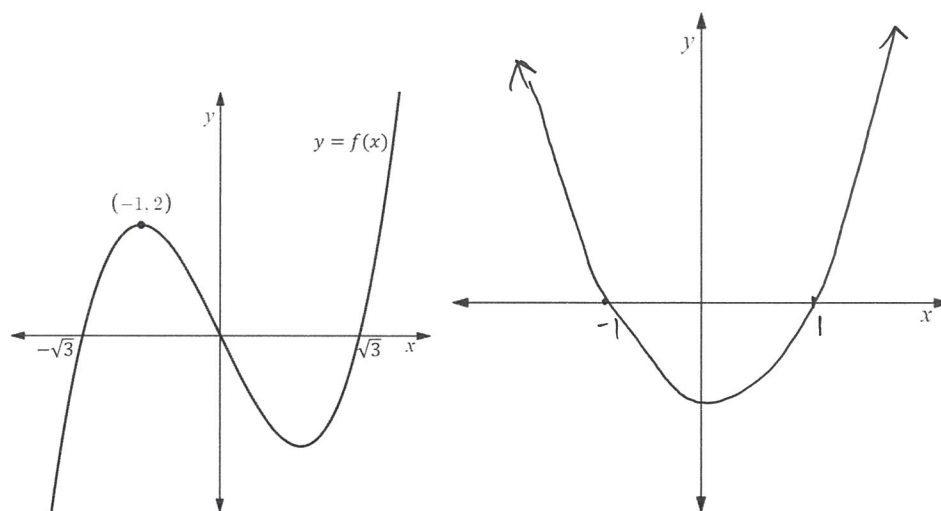
$$y = x + k$$

$$0 = 3 + k$$

$$k = -3$$

$y = x - 3$ will intersect $y = |2x - 6|$
at one point

16.

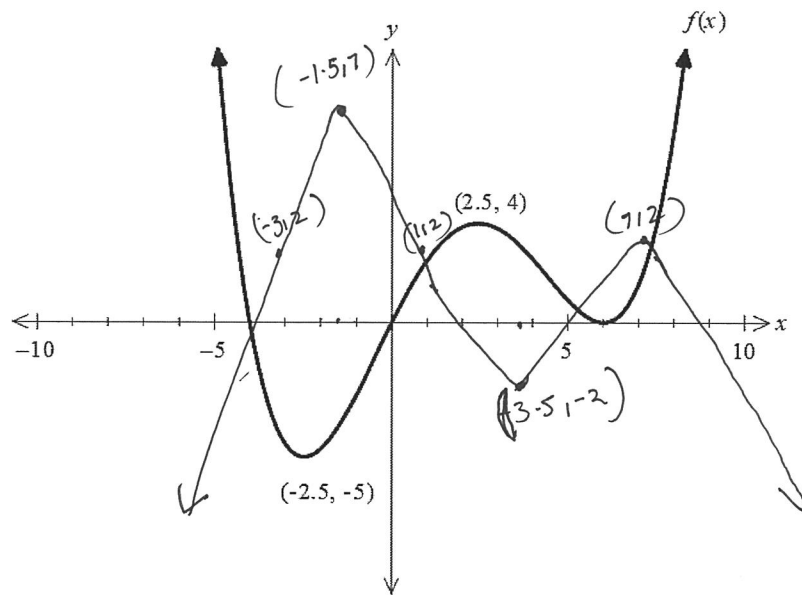


2 MA11-5

The diagram above on the left shows the graph of the curve $y = f(x)$. On the set of axes above on the right, sketch the graph of the curve $y = f'(x)$ showing the shape of the curve and its x intercepts.

18 The graph of $f(x)$ is shown.

3 MA11-2



Graph $y = 2 - f(x-1)$ on the same set of axes showing all essential points.

- 20 The observed brightness of a star can be measured by its energy flux and can be transformed by a logarithmic scale to be more easily measured by an apparent stellar magnitude.

3 MA11-6

It is defined to be $m = -\sqrt[5]{100} \log\left(\frac{f}{f_v}\right)$.

Where m is the magnitude, f is the energy flux of the star and f_v is the reference energy flux.

The star Betelgeuse has a magnitude of 0.45 while the star Deneb has a magnitude of 1.26.

Which star is brighter and by how much brighter is it?

(magnitude of Star Betelgeuse) $m_B = 0.45$ $m_D = 1.26$ (magnitude of star Deneb)
 $f_B = \text{flux of star Betelgeuse}$ $f_D = \text{flux of star Deneb}$

$$1.26 - 0.45 = -\sqrt[5]{100} \log\left(\frac{f_D}{f_r}\right) + \sqrt[5]{100} \log\left(\frac{f_B}{f_r}\right)$$

$$\frac{0.81}{\sqrt[5]{100}} = \log\left(\frac{f_B}{f_r}\right) - \log\left(\frac{f_D}{f_r}\right)$$

$$\frac{0.81}{\sqrt[5]{100}} = \log\left(\frac{f_B}{f_r} \times \frac{f_r}{f_D}\right)$$

$$\frac{0.81}{\sqrt[5]{100}} = \log_{10}\left(\frac{f_B}{f_D}\right)$$

$$10^{\frac{0.81}{\sqrt[5]{100}}} = \frac{f_B}{f_D}$$

$$f_B = 10^{\frac{0.81}{\sqrt[5]{100}}} \times f_D$$

$$\approx 2.1 f_D$$

$\therefore$ Star Betelgeuse is ¹⁴ 2.1 brighter than star Deneb

21. Solve the equation
- $(\log_3 x)^2 = 1$

1 MA11-6

$$\log_3 x = \pm 1$$

$$x = 3^1 \quad x = 3^{-1}$$

$$\boxed{x = 3} \quad \boxed{x = \frac{1}{3}}$$

22. Solve the equation
- $\log_2(x+4) + \log_2(x-2) = 4$
- .

3 MA11-6

$$\log_2 (x+4)(x-2) = 4$$

$$x^2 + 2x - 8 = 2^4$$

$$x^2 + 2x - 24 = 0$$

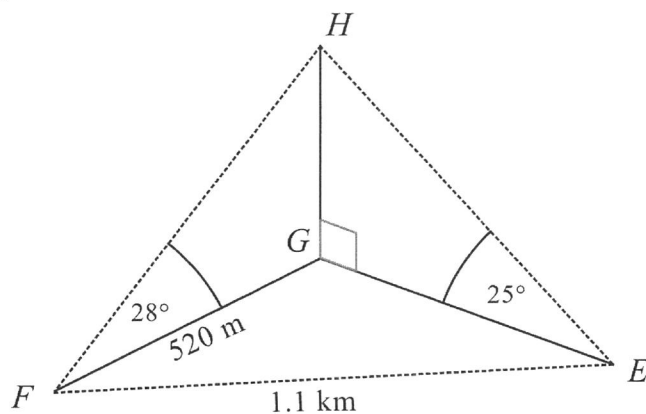
$$(x+6)(x-4) = 0$$

$$x = -6 \quad \boxed{x = 4}$$

$$x > 0$$

Not a solution

- 25 Three points E , F and G lie on a horizontal plain. The distance $EF = 1.1$ km and $FG = 520$ m. A vertical tower HG is constructed so that $\angle HFG = 28^\circ$ and $\angle HEG = 25^\circ$.



NOT TO
SCALE

Calculate the height of the tower HG and hence find the distance EG and the size of $\angle EGF$, rounding each appropriately.

$$\tan 28 = \frac{HG}{520}$$

$$HG = 520 \tan 28 = 276 \text{ m}$$

$$\tan 25 = \frac{HG}{EG}$$

$$EG = \frac{HG}{\tan 25} = \frac{520 \tan 28}{\tan 25}$$

$$EG = 593 \text{ m}$$

$$\cos \angle EGF = \frac{520^2 + 593^2 - 1100^2}{2(520)(593)}$$

$$\angle EGF = 162^\circ$$

26. A function $f(x)$ with range $(0, \infty)$ is such that $f(a) \times f(b) = f(a + b)$ for any real numbers a and b .

Show that $f\left(\frac{a}{2}\right) = \sqrt{f(a)}$ for any real number a .

~~3~~ MA11-1
2

$$f\left(\frac{a}{2}\right) \times f\left(\frac{a}{2}\right) = f\left(\frac{a}{2} + \frac{a}{2}\right)$$

$$\left(f\left(\frac{a}{2}\right)\right)^2 = f(a)$$

$$f\left(\frac{a}{2}\right) = \sqrt{f(a)}$$

TASK 3 YEAR 11 ADVANCED - NSBHS

27. Nick constructs a swimming pool in the shape below.
Nick's pool design is made up of a sector and a semi-circle,
the radius(r) of the sector is equal to the diameter of the semi-circle.

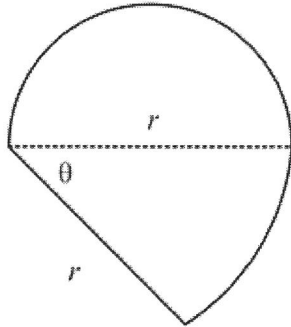


Figure not to scale

- a. Find the perimeter of the pool in terms of r and θ .

2 MA11-3

$$P = \frac{1}{2} \times \pi \times r + r\theta + r$$

$$P = \frac{\pi r}{2} + r\theta + r$$

Nick wants to be able to use this design in several different properties over the next year, so he decides to let the area of the pool equal 1 square unit for future ease in scaling the pool sizes.

- b. Show that, if this is the case, the perimeter can be given by:

$$P = \frac{2}{r} + r \left(1 + \frac{\pi}{4} \right).$$

$$A = \frac{1}{2} \times \pi \times \left(\frac{r}{2} \right)^2 + \frac{1}{2} r^2 \theta$$

$$1 = \frac{\pi r^2}{8} + \frac{4 \times r^2 \theta}{2 \times 4}$$

$$1 = \frac{\pi r^2}{8} + \frac{4 r^2 \theta}{8}$$

$$8 = \pi r^2 + 4 r^2 \theta$$

$$\frac{8 - \pi r^2}{4 r^2} = \theta \quad \theta = \frac{2}{r^2} - \frac{\pi}{4}$$

$$P = \frac{\pi r}{2} + r + r \left(\frac{2}{r^2} - \frac{\pi}{4} \right)$$

$$= \frac{2 \times \pi r}{2 \times 2} + r + \frac{2}{r} - \frac{\pi r}{4}$$

$$= \frac{\pi r}{4} + r + \frac{2}{r}$$

$$= \frac{2}{r} + r \left(1 + \frac{\pi}{4} \right)$$

END OF EXAM